

Выполнил: Жмакин Николай Валерьевич, ису: 339097.

Буду выполнять все под WSL + Docker.

Создал сеть

docker network create demo-network

```
nick@DESKTOP-44JRQK7: ~  
nick@DESKTOP-44JRQK7:~$ docker version  
Client:  
Version:           29.5.2  
API version:       1.54  
Go version:        go1.26.3  
Git commit:        79eb04c  
Built:             Wed May 20 14:37:09 2026  
OS/Arch:           linux/amd64  
Context:           default  
  
Server: Docker Desktop 4.76.0 (228118)  
Engine:  
Version:           29.5.2  
API version:       1.54 (minimum version 1.40)  
Go version:        go1.26.3  
Git commit:        568f755  
Built:             Wed May 20 14:38:09 2026  
OS/Arch:           linux/amd64  
Experimental:      false  
containerd:  
Version:           v2.2.4  
GitCommit:         193637f7ee8ae5f5aa5248f49e7baa3e6164966e  
runc:  
Version:           1.3.5  
GitCommit:         v1.3.5-0-g488fc13e  
docker-init:  
Version:           0.19.0  
GitCommit:         de40ad0  
  
nick@DESKTOP-44JRQK7:~$ docker ps  
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES  
nick@DESKTOP-44JRQK7:~$ docker network create demo-network  
5f5a5d7c35965cb50b809077d7573ba53d9418025bbe7bf5234489682fb79594  
nick@DESKTOP-44JRQK7:~$
```

Создал yml фал в отдельной директории

mkdir -p ~/monitoring/prometheus

cd ~/monitoring/prometheus

touch prometheus.yml

nano prometheus.yml

```
nick@DESKTOP-44JRQK7: ~/monitoring/prometheus  
nick@DESKTOP-44JRQK7:~$ mkdir -p ~/monitoring/prometheus  
nick@DESKTOP-44JRQK7:~$ cd ~/monitoring/prometheus  
nick@DESKTOP-44JRQK7:~/monitoring/prometheus$ touch prometheus.yml  
nick@DESKTOP-44JRQK7:~/monitoring/prometheus$ nano prometheus.yml  
nick@DESKTOP-44JRQK7:~/monitoring/prometheus$
```

```
nick@DESKTOP-44JROK7: ~/monitoring/prometheus
GNU nano 8.7.1 prometheus.yml *
global:
  scrape_interval: 15s

scrape_configs:
  - job_name: 'prometheus'
    static_configs:
      - targets: ['localhost:9090']

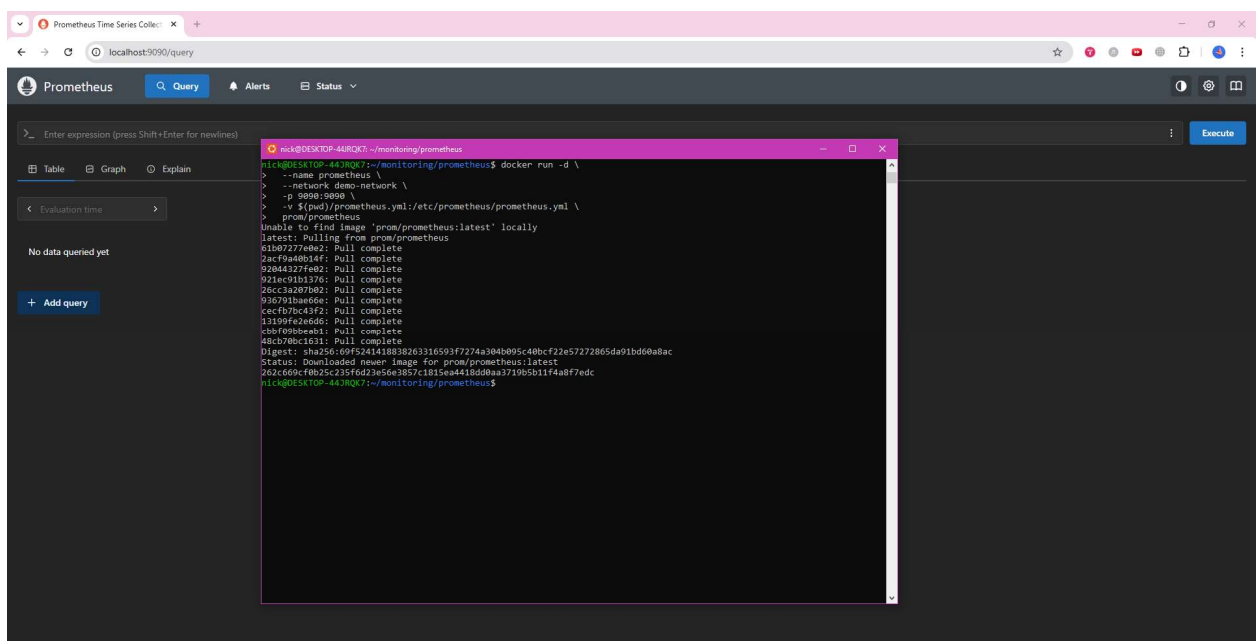
^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo     M-A Set Mark
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line M-E Redo     M-6 Copy
```

Загрузил prometheus

```
docker run -d \
  --name prometheus \
  --network demo-network \
  -p 9090:9090 \
  -v $(pwd)/prometheus.yml:/etc/prometheus/prometheus.yml \
  prom/prometheus
```

```
nick@DESKTOP-44JROK7: ~/monitoring/prometheus
nick@DESKTOP-44JROK7:~/monitoring/prometheus$ docker run -d \
> --name prometheus \
> --network demo-network \
> -p 9090:9090 \
> -v $(pwd)/prometheus.yml:/etc/prometheus/prometheus.yml \
> prom/prometheus
Unable to find image 'prom/prometheus:latest' locally
latest: Pulling from prom/prometheus
61b07277e0e2: Pull complete
2acf9a40b14f: Pull complete
92044327fe02: Pull complete
921ec91b1376: Pull complete
26cc3a207b02: Pull complete
936791bae66e: Pull complete
cecfb7bc43f2: Pull complete
13199fe2e6d6: Pull complete
cbbf09bbeab1: Pull complete
48cb70bc1631: Pull complete
Digest: sha256:69f5241418838263316593f7274a304b095c40bcf22e57272865da91bd60a8ac
Status: Downloaded newer image for prom/prometheus:latest
262c669cf0b25c235f6d23e56e3857c1815ea4418dd0aa3719b5b11f4a8f7edc
nick@DESKTOP-44JROK7:~/monitoring/prometheus$
```

Запустилось!



Загрузил node-exporter, обновил конфиг prometheus

**docker run -d **

**--name node-exporter **

**--network demo-network **

**-p 9100:9100 **

prom/node-exporter

nano prometheus.yml

cat prometheus.yml

```
nick@DESKTOP-44JRQK7: ~/monitoring/prometheus$ docker run -d \
> --name node-exporter \
> --network demo-network \
> -p 9100:9100 \
> prom/node-exporter
Unable to find image 'prom/node-exporter:latest' locally
latest: Pulling from prom/node-exporter
a6680b927350: Pull complete
fe82e83000f4: Pull complete
a2243411ec8e: Pull complete
Digest: sha256:e9cfff4fc67b1818f8c97adb115b9f12c9a54b533de86765d4a0effc01b357205
Status: Downloaded newer image for prom/node-exporter:latest
a65fcc1af29837a6d5201b123fb15b77fca6b06de7bf981539668c55a5a805b4
nick@DESKTOP-44JRQK7:~/monitoring/prometheus$ nano prometheus.yml
nick@DESKTOP-44JRQK7:~/monitoring/prometheus$ cat prometheus.yml
global:
  scrape_interval: 15s

scrape_configs:
  - job_name: 'prometheus'
    static_configs:
      - targets: ['localhost:9090']

  - job_name: 'node'
    static_configs:
      - targets: ['node-exporter:9100']
nick@DESKTOP-44JRQK7:~/monitoring/prometheus$
```

Перезапускаю prometheus
docker restart prometheus

Работает!

The screenshot shows the Prometheus web interface in a browser. The 'Status' tab is selected, showing 'Target health'. There are two target groups: 'node' and 'prometheus'. The 'node' group has one target 'http://node-exporter:9100/metrics' which is 'UP'. The 'prometheus' group has one target 'http://localhost:9090/metrics' which is also 'UP'. Below the targets, a terminal window is open, showing the same commands and output as the first image, including the Docker run command and the contents of prometheus.yml.

Посмотрел метрику

100 - (avg by(instance)
(irate(node_cpu_seconds_total{mode="idle"}[5m])) * 100)

через UI

Добавил и его в конфиг, перезапустил

```
nick@DESKTOP-44JRQK7: ~/monitoring/prometheus
nick@DESKTOP-44JRQK7:~/monitoring/prometheus$ nano prometheus.yml
nick@DESKTOP-44JRQK7:~/monitoring/prometheus$ cat prometheus.yml
global:
  scrape_interval: 15s

scrape_configs:
  - job_name: 'prometheus'
    static_configs:
      - targets: ['localhost:9090']

  - job_name: 'node'
    static_configs:
      - targets: ['node-exporter:9100']

  - job_name: 'app'
    metrics_path: '/actuator/prometheus'
    static_configs:
      - targets: ['app:8080']
nick@DESKTOP-44JRQK7:~/monitoring/prometheus$ docker restart prometheus
prometheus
nick@DESKTOP-44JRQK7:~/monitoring/prometheus$
```

nano prometheus.yml

cat prometheus.yml

docker restart prometheus

Создал файл алерта и добавил его в конфиг prometheus

```
nick@DESKTOP-44JRQK7: ~/monitoring/prometheus
nick@DESKTOP-44JRQK7:~/monitoring/prometheus$ touch alerts.yml
nick@DESKTOP-44JRQK7:~/monitoring/prometheus$ nano alerts.yml
nick@DESKTOP-44JRQK7:~/monitoring/prometheus$ cat alerts.yml
groups:
  - name: cpu_alerts
    rules:
      - alert: HighCPUUsage
        expr: 100 - (avg(irate(node_cpu_seconds_total{mode="idle"}[5m])) * 100) > 80
        for: 1m
        labels:
          severity: warning
        annotations:
          summary: "Высокая нагрузка на CPU"
nick@DESKTOP-44JRQK7:~/monitoring/prometheus$ nano prometheus.yml
nick@DESKTOP-44JRQK7:~/monitoring/prometheus$ cat prometheus.yml
global:
  scrape_interval: 15s

scrape_configs:
  - job_name: 'prometheus'
    static_configs:
      - targets: ['localhost:9090']

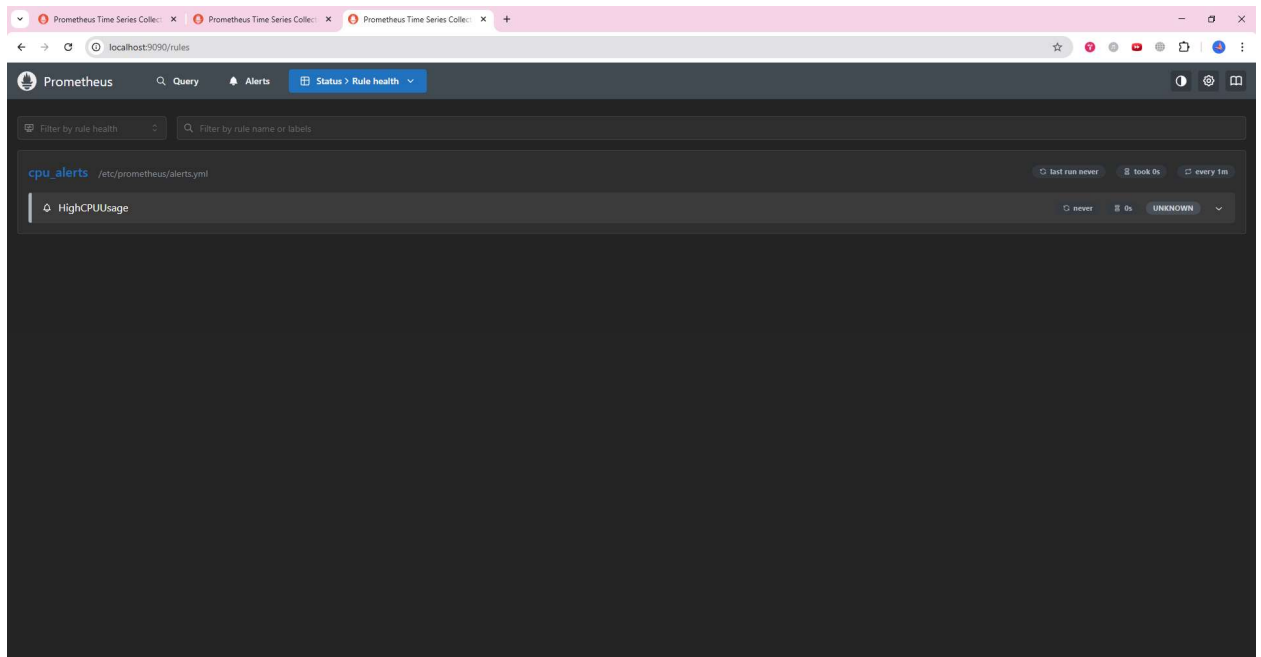
  - job_name: 'node'
    static_configs:
      - targets: ['node-exporter:9100']

  - job_name: 'app'
    metrics_path: '/actuator/prometheus'
    static_configs:
      - targets: ['app:8080']

rule_files:
  - "alerts.yml"
nick@DESKTOP-44JRQK7:~/monitoring/prometheus$ docker restart prometheus
prometheus
nick@DESKTOP-44JRQK7:~/monitoring/prometheus$
```

После этого пересоздал контейнер, т.к. этот файл алертов не был подключен к докеру.

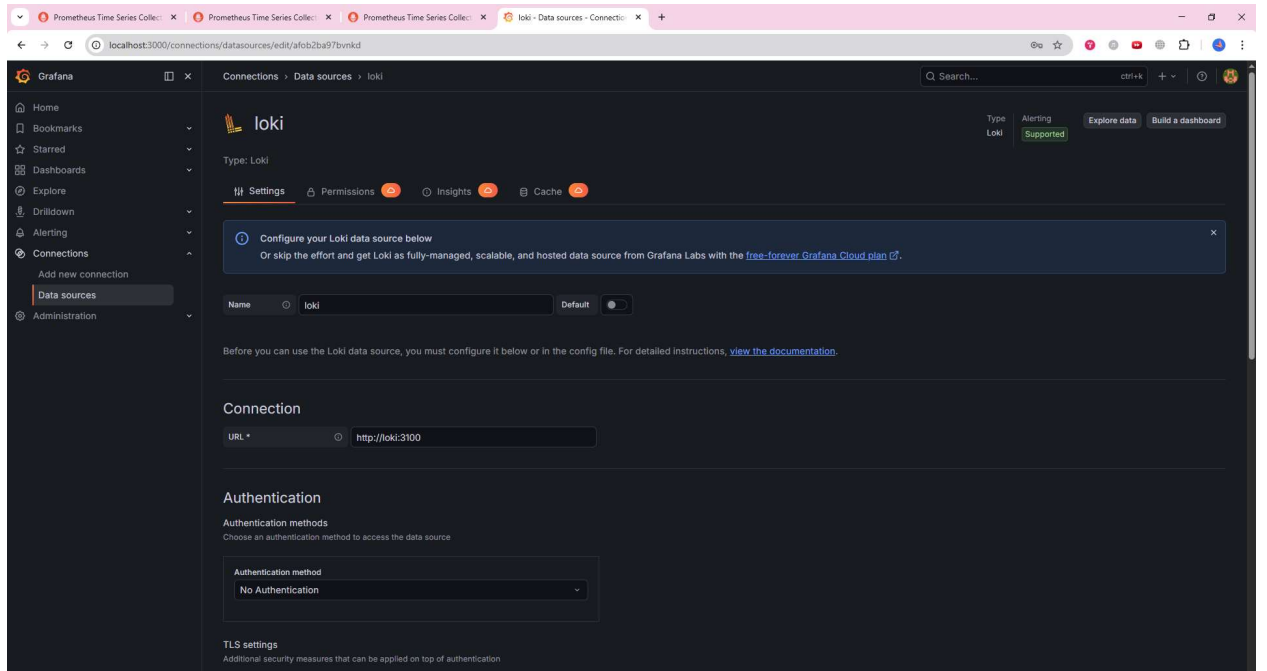
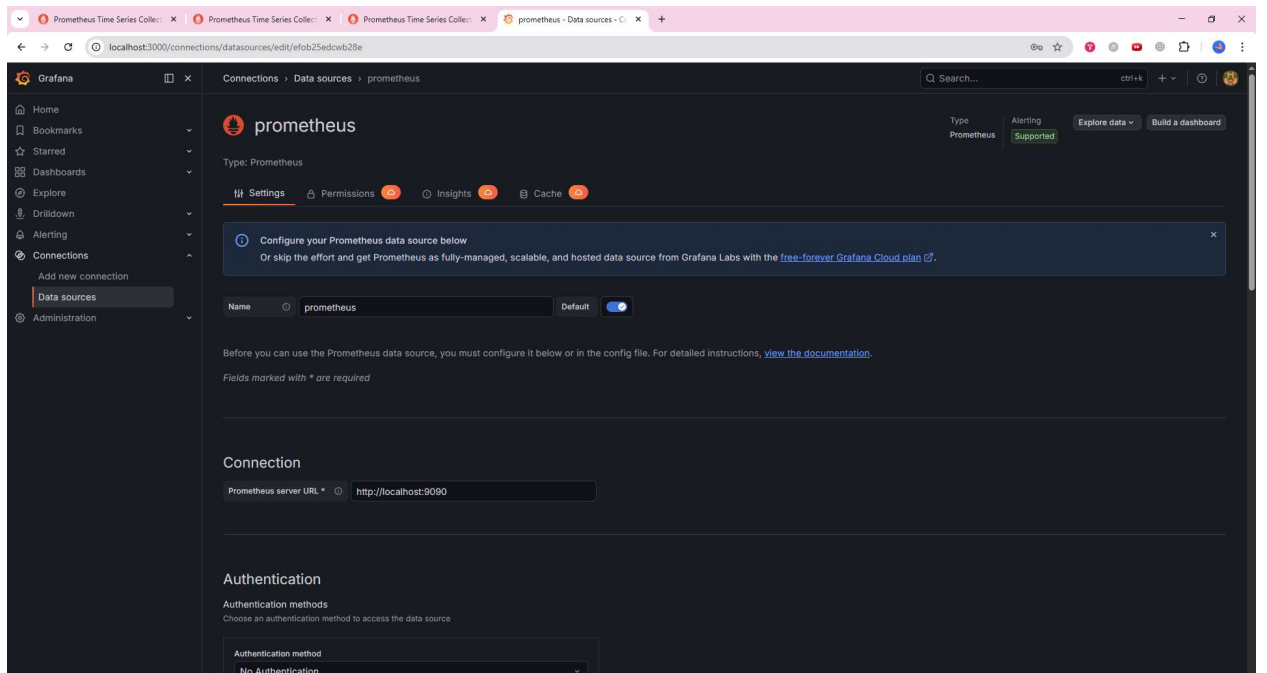
После этого появился алерт в списке.

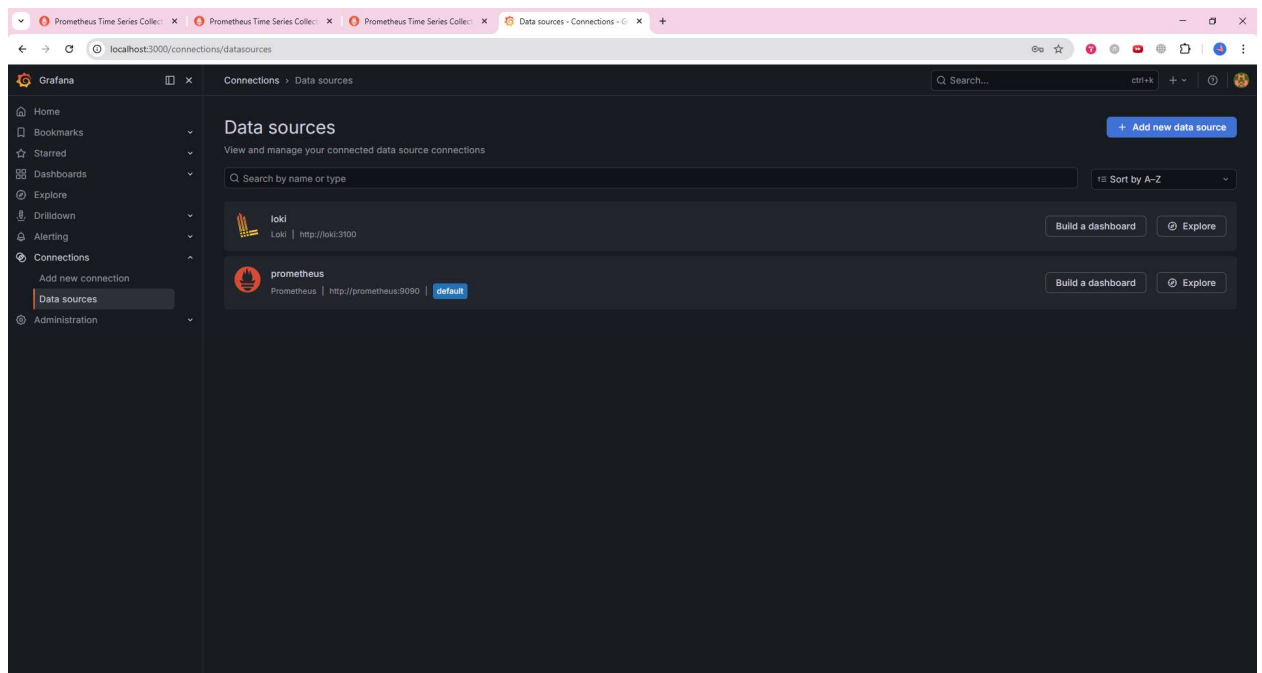


Выделил в отдельную папку и туда залил все конфиги для compose

```
nick@DESKTOP-44JRK7: ~/monitoring/prometheus/monitoring-stack
nick@DESKTOP-44JRK7:~/monitoring/prometheus$ mkdir monitoring-stack
nick@DESKTOP-44JRK7:~/monitoring/prometheus$ cd monitoring-stack/
nick@DESKTOP-44JRK7:~/monitoring/prometheus/monitoring-stack$ nano docker-compose.yml
nick@DESKTOP-44JRK7:~/monitoring/prometheus/monitoring-stack$ nano loki-config.yaml
nick@DESKTOP-44JRK7:~/monitoring/prometheus/monitoring-stack$ nano promtail-config.yaml
nick@DESKTOP-44JRK7:~/monitoring/prometheus/monitoring-stack$ docker compose up -d
WARN[0000] /home/nick/monitoring/prometheus/monitoring-stack/docker-compose.yml: the attribute `version` is obsolete, it
will be ignored, please remove it to avoid potential confusion
[+] up 41/41
  Image grafana/loki:latest      Pulled      19.4s
  Image grafana/promtail:latest  Pulled      19.4s
  Image grafana/grafana:latest   Pulled      61.0s
  Volume monitoring-stack_grafana_data Created      0.0s
  Volume monitoring-stack_loki_data Created      0.0s
  Container loki                 Started     15.9s
  Container promtail            Started     15.7s
  Container grafana              Started     15.9s
nick@DESKTOP-44JRK7:~/monitoring/prometheus/monitoring-stack$
```

Добавил источник





Добавил различные метрики на дашборд

